

Pharmacology of Ganglionic and Neuromuscular Nicotinic Receptor Antagonists

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I am available to groups and individuals for pharmacology help and discussions by appointment.

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After completing the preparation materials, students should be able to:

1. PK properties → onset and duration of action

Relate the pharmacokinetic properties of the **ganglion blocking drugs** and the **neuromuscular blocking agents (NMBAs)** to their onset and duration of action in patients with full hepatic and renal function and those with impaired hepatic and renal function by giving specific examples.

2. Structure-activity relationship → Site of action and Mechanism of action

Explain how the structural features of the **ganglion blocking drugs** and the **NMBAs** determine relative specificity for neuronal or neuromuscular nicotinic receptors, respectively.

3. Site of action / Mechanism of action → Therapeutic uses and Adverse effects

- **Ganglion blockers:** Predict the therapeutic and adverse effects of ganglion nicotinic receptor blockade on organ systems exemplified by mecamylamine.
- **NMBAs:** Differentiate the actions and effects of depolarizing neuromuscular blockade by succinylcholine and the nondepolarizing NMBAs.

4. NMBA reversal

Explain the mechanisms of the cholinesterase inhibitors and sugamadex for reversing neuromuscular block and their potential adverse effects including specific examples.

5. Mechanisms and effects → Drug-drug and drug-disease interactions

Explain the mechanisms and effects of drug-drug and drug-disease interactions of the ganglion blockers and NMBAs.

Preparation Materials

Required

- **ScholarRx Bricks | Practice Questions** (not graded)

Supplemental resources if you wish to use them (they are not required):

- Video lecture | Dr. Goldstein's Word handout | Guided Reading Questions
- Links to a textbook and pharmacology review books are provided on the next slide.

STUDY TIPS:

- I provide several learning options with links so your are aware of them.
- ***You do not need to study all of them. Use the resources that you like to use for learning.***
- Work through the GUIDED READING QUESTIONS with pen/pencil and paper.

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

- Practice questions are intended to help you integrate and apply pharmacology concepts in the context of case vignettes and will help you become familiar with board-style of exam questions.

Links to resources

Scholar Rx Bricks: Pharmacology of Antimuscarinic Drugs

Bricks: **General Pharmacology > Autonomic Drugs > Drugs that Alter Parasympathetic Tone** Parasympatholytics > Antimuscarinic Drugs <https://exchange.scholarrx.com/brick/muscarinic-antagonists>

Pharmacology of Ganglion Blockers and Neuromuscular Blockers

Bricks: **Neurology and Special Senses > Disorders of the Nervous System and Special Senses > Pain Mechanisms and Treatment** Principles of Anesthesia > Neuromuscular Blocking Drugs https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_NEU0100

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2023; Chapter 8: Cholinoceptor-Blocking Drugs > Ganglionic-Blocking Drugs <https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com/content.aspx?bookid=3382§ionid=281747661#1204136483>

Katzung's Chapter 27: Skeletal Muscle Relaxants; <https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com/content.aspx?bookid=3382§ionid=281751187#1206641390>

REVIEW BOOKS HAVE PRACTICE QUESTIONS

LWW Health Library, Premium Basic Sciences: Lippincott's Illustrated Reviews: Pharmacology; 8e, 2023; Chapter 5: Cholinergic Antagonists <https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary-com/content.aspx?sectionid=253325538&bookid=3222>

Katzung's Pharmacology: Examination & Board Review, 14e, 2024; Chapter 8: Cholinoceptor-Blocking Drugs & Cholinesterase Regenerators <https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com/content.aspx?bookid=3461§ionid=285590012>
Chapter 27: Skeletal Muscle Relaxants > Neuromuscular Blocking Agents <https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com/content.aspx?bookid=3461§ionid=285594147>

The following key points are intended help the student grasp the main ideas and concepts by organizing and summarizing the “need to know” information. Key points may also serve as a concise reference to facilitate review.

Key points: PK properties → onset and duration of action

Ganglion blocker

Mecamylamine	oral, complete absorption, crosses BBB, urine/unchanged drug, onset 30-120 min; duration 6->12 h; dosing 2-4x daily based on BP response
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NMBAs: all IV; polar – do not cross BBB; onset and duration depend on age and dose

Succinylcholine	IV/M; rapidly hydrolyzed by plasma cholinesterase to inactive metabolites. rapid onset (IV <60 sec; IM 2-3 min), short duration (IV <8 min; IM 10-20 min) Patients with deficient or absent plasma cholinesterase (rare) have increased duration of neuromuscular block, which could lead to prolonged apnea.
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Benzyloisoquinolines

Atracurium	ester hydrolysis and Hofmann elimination, laudanosine metabolite onset in minutes, recovery begins in ~30-45 min
Cisatracurium	Hofmann elimination, onset in minutes, recovery begins in ~30-45 min

Aminosteroids

Rocuronium	biliary/feces excretion, onset 1-2 min, recovery 20-120 min
Vecuronium	biliary and renal excretion, onset 2-4 min, recovery begins in ~20-40 min

Key points:

Structure-activity relationship → Site of action and Mechanism of Action

Ganglion blockers are synthetic amines with structural characteristics similar to acetylcholine. They are competitive inhibitors of neuronal nicotinic acetylcholine receptors (nAChRs). Ganglion blockers have lower affinity for neuromuscular nAChRs.

- Mecamylamine is the only one currently in clinical use. It's bulky bicyclic structure and secondary amine are necessary for interaction with neuronal nAChRs.
- Ganglion blockers block the actions of acetylcholine at nicotinic receptors – parasympathetic and sympathetic autonomic ganglia and the adrenal medulla – which blocks the entire output of the ANS. Mecamylamine is lipophilic, crosses the blood-brain barrier and also inhibits central nAChRs.

Neuromuscular blocking agents (NMBAs) also have structural characteristics similar to acetylcholine. They all have one or two quaternary ammoniums. NMBA's currently in clinical use have lower affinity for neuronal nAChRs.

- Succinylcholine is a **depolarizing** NMBA with a linear structure consisting of two acetylcholine molecules. Succinylcholine initially opens the channel, causing depolarization of the motor end plate and transient contractions. Succinylcholine is not metabolized in the synapse so the membranes remain depolarized and unresponsive to subsequent impulses = depolarizing blockade.
- The **nondepolarizing** NMBAs have a bulky, semi-rigid ring structure. They are subcategorized according their structures – isoquinolines and aminosteroids. These drugs are competitive inhibitors of acetylcholine. In larger doses they can enter the channel pore, producing a more intense blockade.

Key points:

Mechanism of action / Site of action → Therapeutic uses and Adverse effects

Ganglion blockers:

Mecamylamine is used in the treatment of moderate to severe hypertension and uncomplicated malignant hypertension. Many drugs with more selective actions and fewer side effects are available.

The adverse effects are related to the physiologic responses. The physiologic responses are related to:

- 1) the responses of the effector organs to autonomic nerve impulse, and
- 2) the division of the autonomic nervous system exerting the dominant tone at the various organs.
- 3) the homeostatic reflexes that regulate autonomic tone (activity) are *absent*.

When drugs block the ganglion nAChRs, the prevailing (dominant) parasympathetic or sympathetic tone is reversed.

- Responses are complex and mostly unpredictable.
- All organs of the autonomic nervous system are affected. Please see the table on slide #16.
- Mecamylamine causes severe orthostatic hypotension (dizziness, lightheadedness, syncope), constipation, and urinary retention.

Key points:

Mechanism of action / Site of action → Therapeutic uses and Adverse effects

Neuromuscular nicotinic receptor blockers:

- NMBA block transmission of the end plate potential, resulting in skeletal muscle paralysis (muscle relaxation), which facilitates endotracheal intubation and surgical procedures. They may be used to assist mechanical ventilation in the intubated patient.
- The respiratory muscles (intercostals and diaphragm) are the last to relax and the first to recover.
- Succinylcholine is useful for rapid endotracheal intubation and procedures requiring brief muscle relaxation, such as electroconvulsive therapy, because of its rapid onset of action and short duration of action.
- Adjunctive use of nondepolarizing NMBAs provides adequate muscle relaxation for all types of surgical procedures without the cardiorespiratory depressant effects produced by deep anesthesia.
Prior to the introduction of NMBAs, only deep anesthesia with inhaled anesthetics that caused profound depressant effects on cardiovascular and respiratory systems would produce adequate muscle relaxation.
- Succinylcholine has numerous potential adverse effects. It can cause ventricular dysrhythmia, cardiac arrest, and death from hyperkalemia in pediatric patients who were subsequently found to have undiagnosed skeletal muscular dystrophy. The major adverse effect of nondepolarizing NMBAs is prolonged paralysis, respiratory depression, and apnea.
- Age: Neonates may be more sensitive to the effects. Older patients may have reduced clearance.
- Hypersensitivity reactions occur rarely.

NMBA reversal agents

Cholinesterase inhibitors: Neostigmine or Pyridostigmine

- Increase the concentration of ACh in the neuromuscular junction → ACh outcompetes the NMBA at the nAChR → prompt reversal of NMBA-induced skeletal muscle paralysis
- They reverse nondepolarizing neuromuscular blockade by the isoquinolines and aminosteroids.
- Large doses of cholinesterase inhibitor when neuromuscular block is minimal (near recovery) can result in neuromuscular dysfunction due to excessive acetylcholine in the neuromuscular junction.

Sugammadex – rapid reversal:

- Chemical chelator (γ -cyclodextrin) forms a complex in plasma with the aminosteroids, rocuronium and vecuronium (but not the benzylisoquinolines), in a 1:1 ratio. The complex is eliminated in the urine.

Adverse effects:

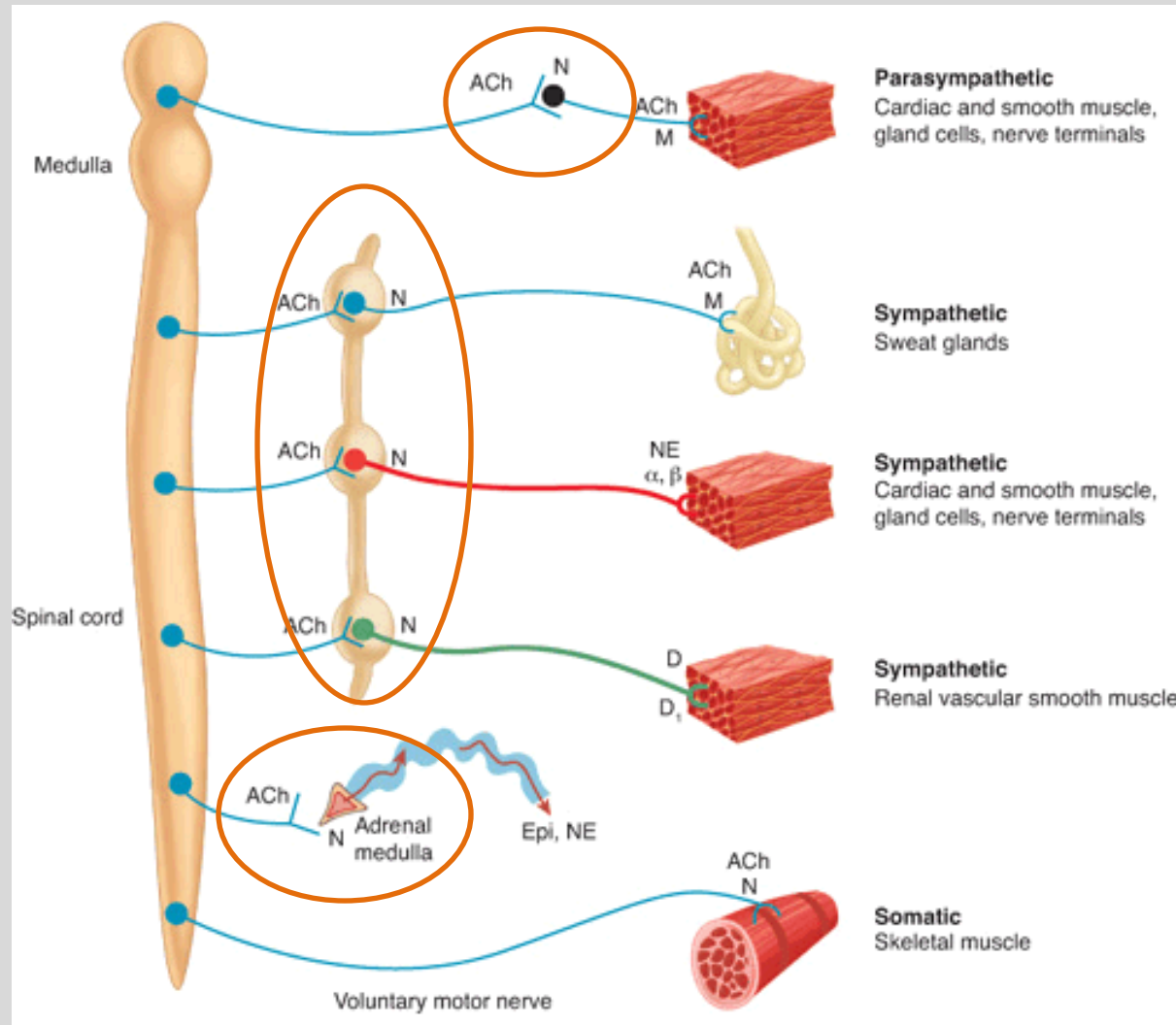
- May cause bradycardia and ECG abnormalities (monitor during drug administration)
- Serious hypersensitivity reactions / anaphylaxis reported (uncommon)
- Increases in coagulation parameters; clinically significant bleeding not reported
- Sugammadex may decrease plasma concentrations of hormonal contraceptive by steroid encapsulation.

Succinylcholine reversal by diffusion:

- Spontaneous reversal as SCh diffuses away from synapse
- Cholinesterase inhibitors augment depolarizing block by increasing ACh concentration in the synapse, which contributes to the depolarizing block.

Key points: Mechanisms and effects → Drug-drug and drug-disease interactions

Interaction with ganglion blocker	Effect
Drugs acting at autonomic receptors on effector organs	Enhance or reverse BP lowering effect
Drugs with the same adverse effects	Enhance CNS and peripheral side effects
Interaction with NMBAs	Effect
Volatile halogenated anesthetics	Enhance NM blockade Desirable – Adjunctive NMBA allows lower anesthesia concentration.
Aminoglycoside antibiotics Fluoroquinolone antibiotics Tetracyclines	Potentiate NM blockade: Interference with ACh release in the NM junction and/or interfering with receptor function → prolonged muscle weakness and respiratory depression
Many other drugs may cause significant adverse events	Mechanisms: Potentiation or inhibition of effect Enhance respiratory depressant effects Enhance other adverse effects Contribute to electrolyte imbalance
Neuromuscular autoimmune disease Myasthenia gravis Lambert-Eaton myasthenic syndrome	Decreased NM nicotinic receptors (MG) Decreased ACh release (LEMS)  potentiate NMBAs' effects.



Ganglion Blockers

- drugs that block neuronal nicotinic acetylcholine receptors (nAChRs)

* Mecamylamine

- Trimethaphan (historical mention)
- Hexamethonium (research use)

Ganglion blockers are rarely used therapeutically due to myriad side effects.

➤ If ganglion blockers are not often used, why are they included in pharmacology lectures?

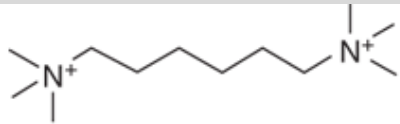
Understanding the effects of ganglion blockers requires a deep knowledge of the autonomic nervous system, including ANS regulation of heart rate, blood pressure, gastrointestinal motility, and ocular dynamics.

Grasping the concepts means students can appreciate how ganglion blockers influence various physiological processes and manage the potential side effects and clinical implications effectively.

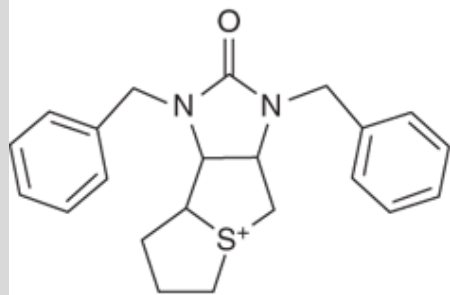
Ganglion Blocking Drugs:

NON-depolarizing neuronal nAChR block of receptors → block ganglionic transmission of **parasympathetic** and **sympathetic** autonomic ganglia and adrenal medulla.

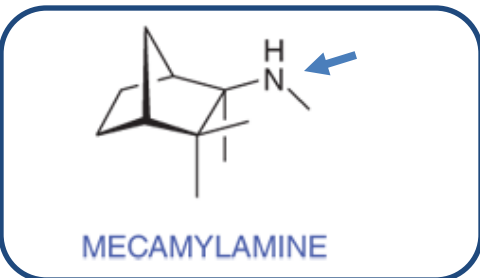
Compare to nicotine, which causes dose-dependent DEpolarizing block of all nicotinic receptors.



HEXAMETHONIUM (C6)



TRIMETHAPHAN



MECAMYLAMINE

Mecamylamine:

Mechanism: Channel plug – blocks the open channel pore of the nAChR complex.

The channel closes around the drug when the agonist unbinds trapping the drug molecule inside. Persistent block cannot be overcome by adding more agonist.

Structure-activity relationship: uncharged and lipophilic

- Rigid bicyclic structure fits the binding site of nAChRs.
- The secondary amine (arrow) is critical for interaction with the acetylcholine receptor by forming hydrogen bonds.
- Lower affinity for neuromuscular nAChRs.

Pharmacokinetics

- Oral, excreted unchanged in urine, duration 6-12h
- Mecamylamine crosses the blood-brain barrier and *inhibits central nAChRs*.
- Weak base: Acidic urine increases rate of excretion and may decrease therapeutic effect.

- Trimethaphan: competitive antagonist, does not cross BBB
- Hexamethonium: blocks the nicotinic ion channel after it is open

Mecamylamine:
lipophilic



distributes to CNS

Responses to treatment with ganglion blockers are complex and largely unpredictable. (Details on the next slide.)

Ganglion blockers...

- suppress outflow from nAChRs of both parasympathetic and sympathetic ganglia and adrenal medulla
- **suppress of all autonomic outflow**

Ganglion blockers...

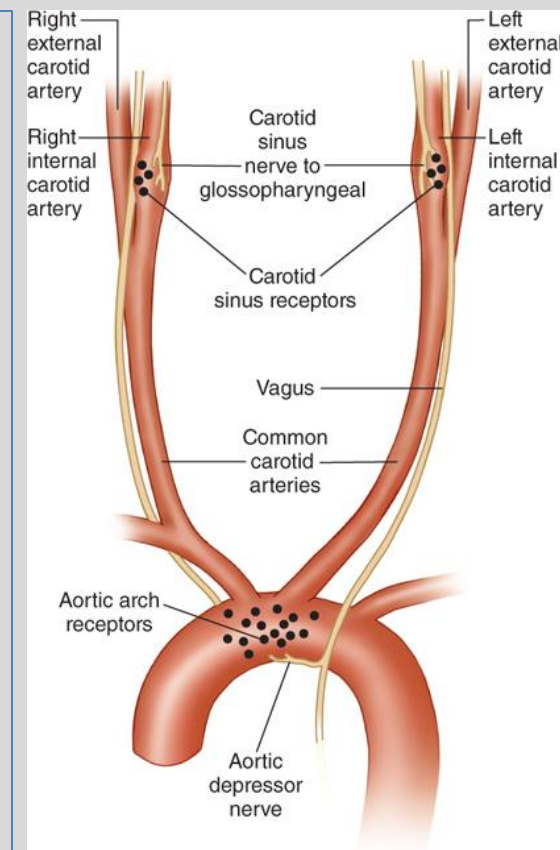
- suppress homeostatic reflexes that normally moderate autonomic responses
- **suppress normal cardiac, vascular, neurologic, muscle, and neurohumoral reflexes**

Mechanism of baroreceptor signaling (which is suppressed by ganglion blockers):

- The ganglia are relay points in the ANS.
- They transmit signals from the baroreceptors to the brain and back to the effector organs.
- When they are disrupted, the signaling is prevented.

Examples:

- Rapid fall in blood pressure sensed by baroreceptors sends signals to the medulla oblongata → the sympathetic nervous system is disinhibited (is increased) → stimulates heart rate and vasoconstriction to increase blood pressure.
- Rapid rise in blood pressure sensed by baroreceptors sends signals to the medulla oblongata → parasympathetic nervous system activity increases → slows the heart rate and stimulates vasodilation to lower blood pressure.



The dominant resting tone to the various effector sites is established by the parasympathetic or sympathetic nervous system (center columns on the table). Even at rest, the sympathetic nervous system maintains a basal level of activity.

When drugs block the ganglion nAChRs, the prevailing (dominant) parasympathetic or sympathetic tone is reversed (left column on the table).

Site	Dominant tone	Physiologic effect	Effects of ANS suppression by ganglion blockade
Arterioles	Sympathetic	Vasoconstriction	Dilation: ↑peripheral blood flow; ↓peripheral vascular resistance; abrupt drop in BP, hypotension
Veins	Sympathetic	Vasoconstriction	Dilation: peripheral pooling of blood; ↓venous return; ↓cardiac output → severe orthostatic hypotension and syncope
Heart	Parasympathetic	Slowed heart rate	Tachycardia
Iris	Parasympathetic	Miosis	Mydriasis (mild)
Ciliary muscle	Parasympathetic	Contraction Adjustment of lens for accommodation	Cycloplegia (ciliary muscle is prevented from contracting) Paralysis of ciliary muscle → loss of accommodation.
GI tract	Parasympathetic	Increased tone, motility, and secretions	Motility can be profoundly affected, causing marked constipation. Secretion of gastric acid is reduced but not enough to treat peptic ulcer.
Urinary bladder	Parasympathetic	Promotes urination	Urinary retention in males with prostatic hyperplasia
Genital tract	Parasympathetic and sympathetic	PNS: erection SNS: ejaculation	Sexual impairment
Salivary glands	Parasympathetic	Copius, watery secretions	Xerostomia
Sweat glands	Sympathetic (cholinergic)	Diaphoresis	Anhidrosis (inability to sweat normally)

Therapeutic Uses

- Chronic hypertension
- Malignant hypertension (uncomplicated)

Lack of selectivity causes a broad range of undesirable effects.

- **Mecamylamine causes orthostatic hypotension** (absent baroreceptor reflexes)
→ reduced blood flow to brain → dizziness, lightheadedness, syncope

Drug interactions *High potential*

- Patients receiving ganglionic blockers are fully responsive to autonomic drugs acting on autonomic receptors at the effector organs: mAChRs and at α ARs and β ARs → exaggerated or reversed responses because the homeostatic reflexes are absent.
- Adverse effects may be enhanced by other drugs producing the same side effects.

<i>Contraindications</i>	Mechanism
Coronary insufficiency	Reduced SNS regulation of heart function and blood vessel tone
Recent myocardial infarction	
Pyloric stenosis	Gastric stasis, worsening nausea, vomiting
Glaucoma	Cycloplegia → ↓ aqueous humor outflow → ↑ intraocular pressure
Uremia	Reduced blood flow to kidney → kidney impairment → waste products accumulate
Unreliable, uncooperative patients	Mecamylamine dosing can be complicated, therapy must be carefully monitored, and the drugs cause orthostatic hypotension.

Check your knowledge.

1. **What is the overall effect of ganglion blockade by mecamylamine?**
2. **What are the mechanism of orthostatic hypotension and symptoms caused by mecamylamine?**
3. **Why should mecamylamine be prescribed only for patients who are reliable and cooperative with the therapy?**
4. **Since mecamylamine is rarely used, why is it important for students to learn about it (other than because it might appear on a board exam?)**

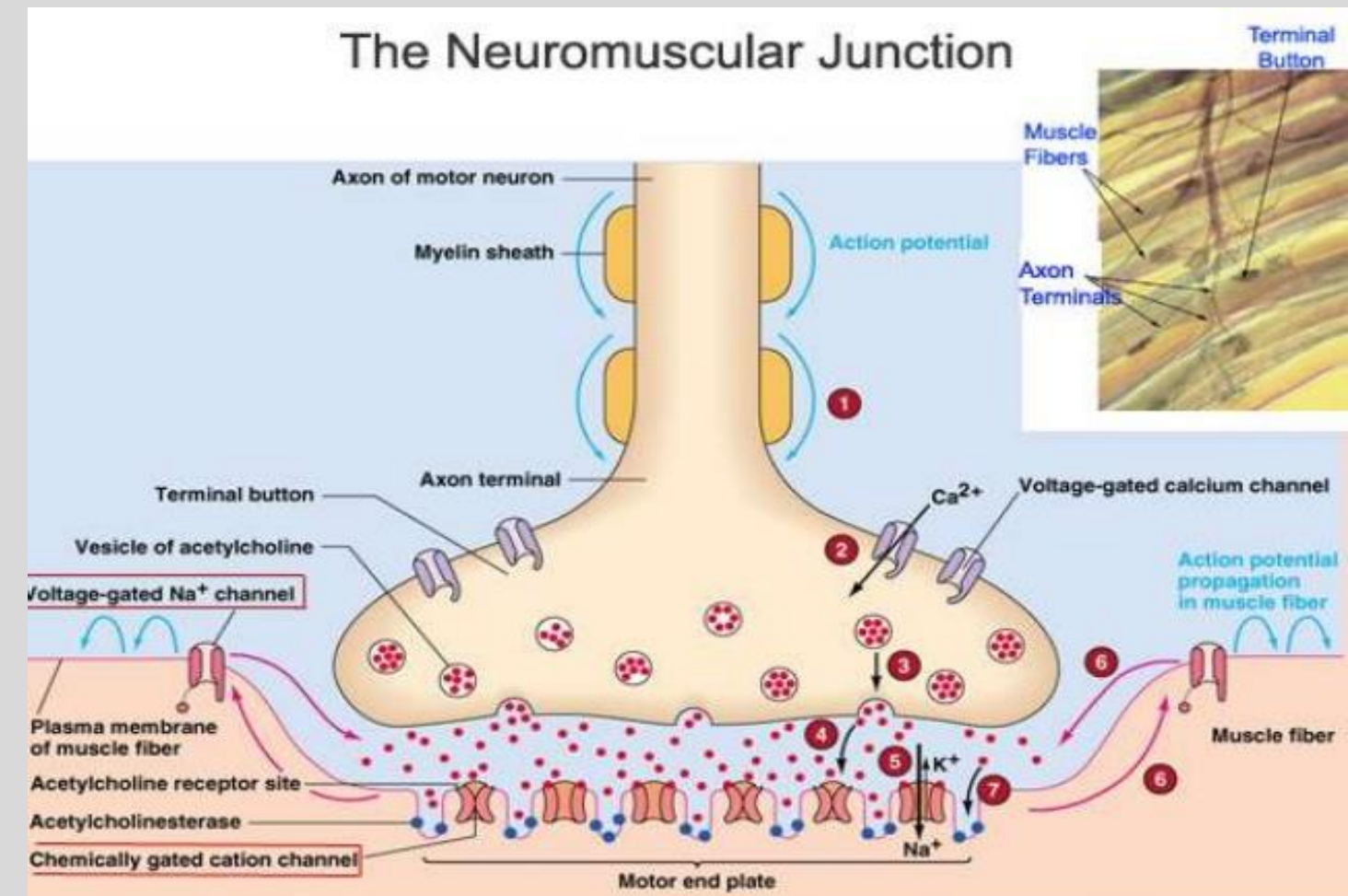
Neuromuscular Blocking Agents (NMBAs)

Nondepolarizing NMBAs

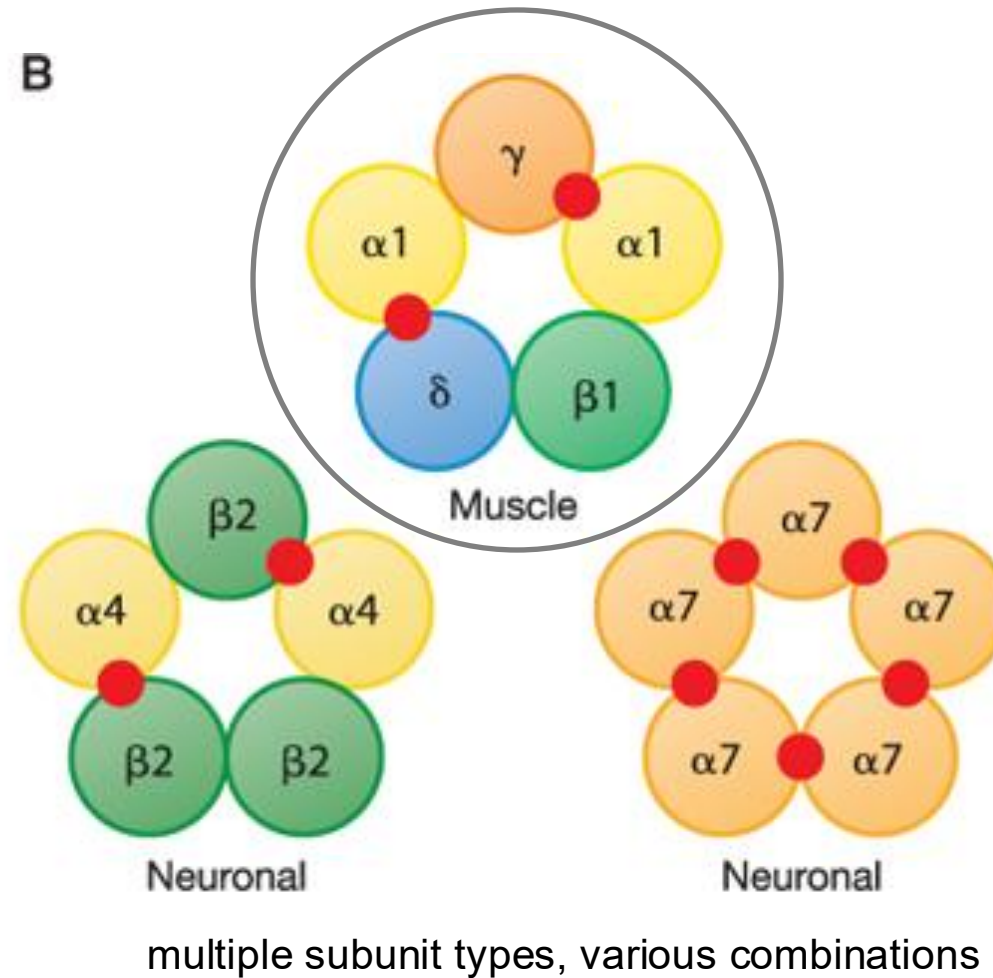
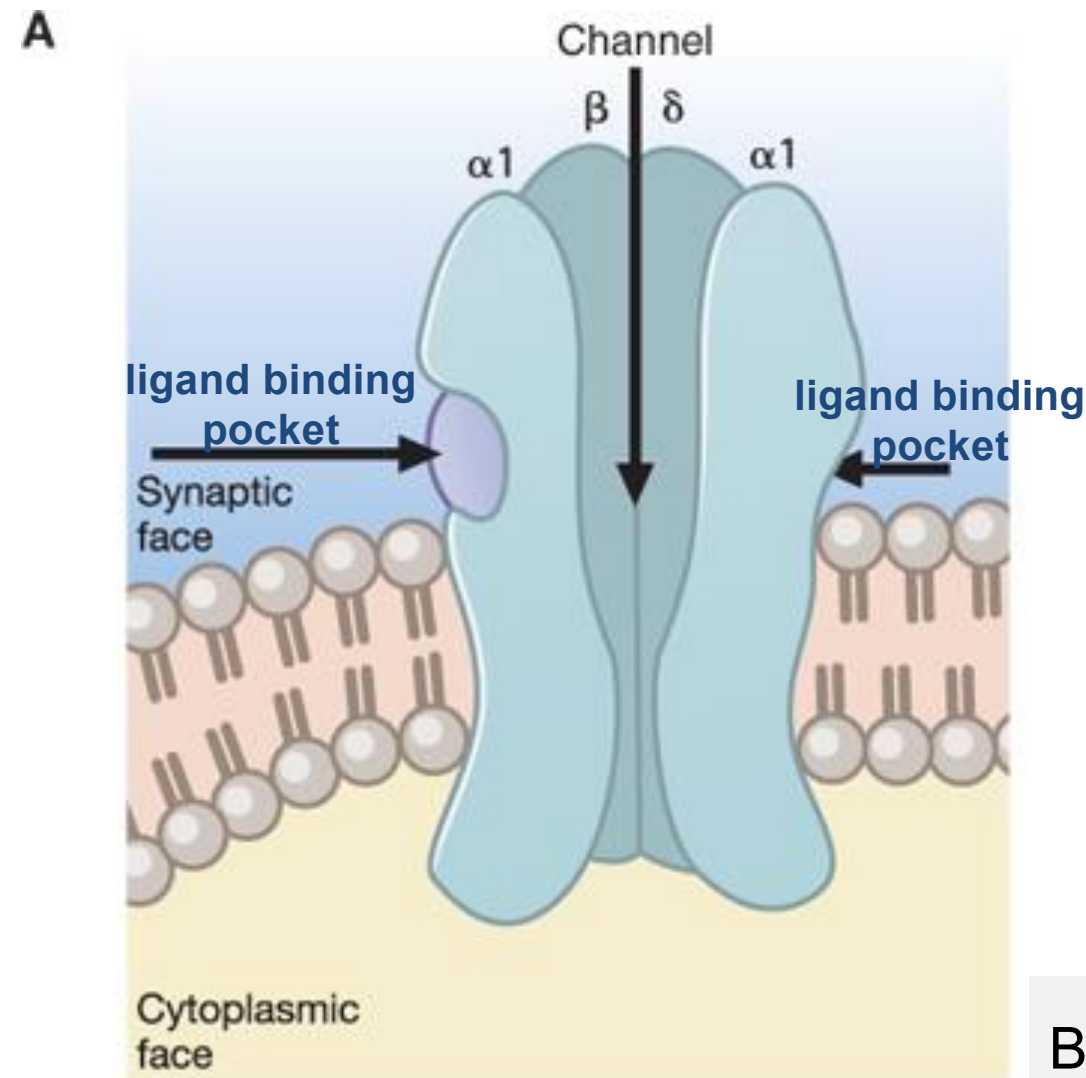
Isoquinolines	Aminosteroids
*Atracurium	*Rocuronium
*Cisatracurium	*Vecuronium
d-Tubocurarine	Pancuronium

Depolarizing NMBA

*Succinylcholine



Structure and subunit arrangement and of neuromuscular nAChRs



A: Five subunits are arranged symmetrically around a central pore.

Muscle-type receptors in the neuromuscular junction typically consist of two $\alpha 1$ subunits, one β subunit, one δ , and one γ (fetus) or ϵ (adult) subunit.

The subunits surround an internal channel with an outer vestibule and its constriction located deep in the membrane bilayer region.

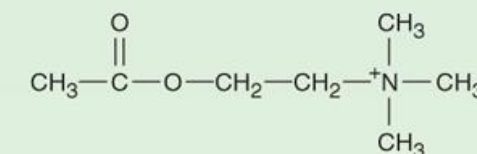
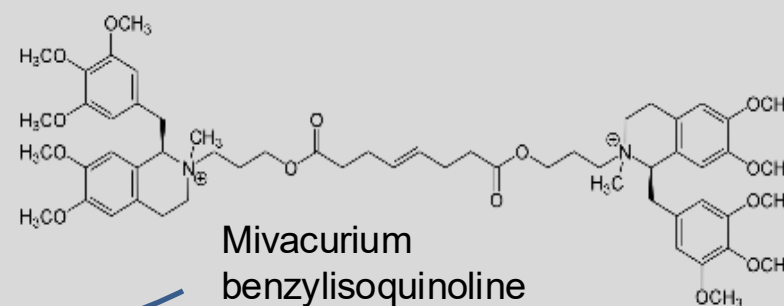
B: The ligand binding sites (*red circles*) are located at the interfaces between subunits: $\alpha 1$ - γ/ϵ and $\alpha 1$ - δ subunits of the neuromuscular nAChRs.

Two ACh molecules must bind the receptors → conformational change → ion channel opens → ions flow through to initiate response.

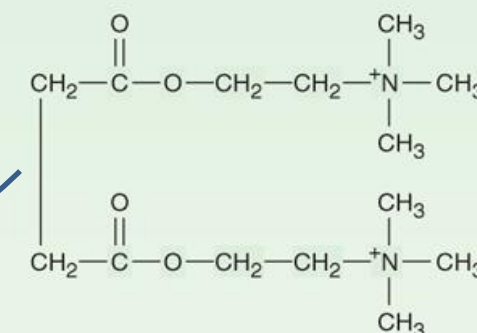
Source: Laurence L. Brunton, Randa Hilal-Dandan, Björn C. Goodman & Gilman's: The Pharmacological Basis of Therapeutics, Thirteenth Edition: Copyright © McGraw-Hill Education. All rights reserved.

NMBAs Structure-Activity Relationship

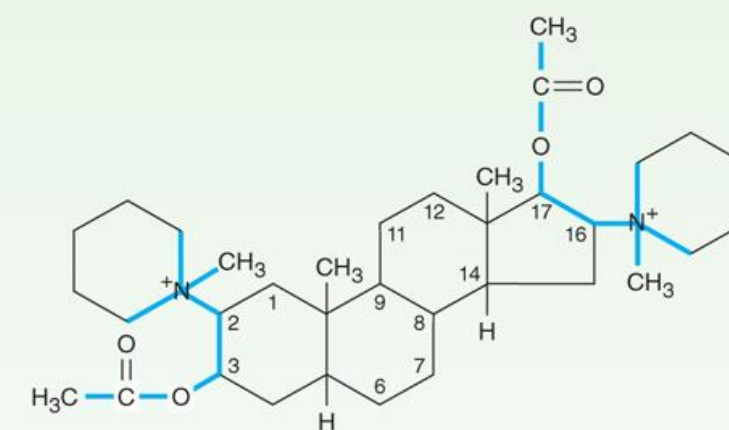
- Structurally similar to acetylcholine
- All NMBAs have 1 or 2 quaternary ammonium groups
 - polar molecules; do not cross into CNS
- Succinylcholine (depolarizing blockade)
 - 2 acetylcholine molecules linked together in a linear, flexible structure
- Isoquinolines and Aminosteroids (nondepolarizing blockade)
 - bulky, semi-rigid molecules



Acetylcholine



Succinylcholine



Pancuronium
aminosteroid

Katzung Figure 27-3. Pancuronium: 2 Acetylcholine-like fragments are outlined

Nondepolarizing NMBAs

Isoquinolines	Aminosteroids
*Atracurium	*Rocuronium
*Cisatracurium	*Vecuronium
<i>d</i> -Tubocurarine	Pancuronium

*All CLASS ACTIONS and EFFECTS

D-Tubocurarine is an alkaloid of American tropical plants.

It is the prototype of the nondepolarizing NMBAs but no longer in clinical use.

It is long-acting, causes ganglion blockade, antimuscarinic effects, and histamine release.

Drug	Elimination	Onset	Duration (95% recovery)	Potency*
BENZYLISOQUINOLINES Intermediate-acting:				
Atracurium	Ester hydrolysis (hepatic) and spontaneous Hofmann elimination (nonenzymatic, independent of liver/kidney function) · Laudanosine byproduct, CNS stimulant · Longer half-life than atracurium's half-life · May accumulate with long infusions (days)	2-3 min	20-35 min (90 min)	1.5
Cisatracurium cis-isomer of atracurium	Hofmann elimination	2-3 min	25-44 (~30-90)	1.5

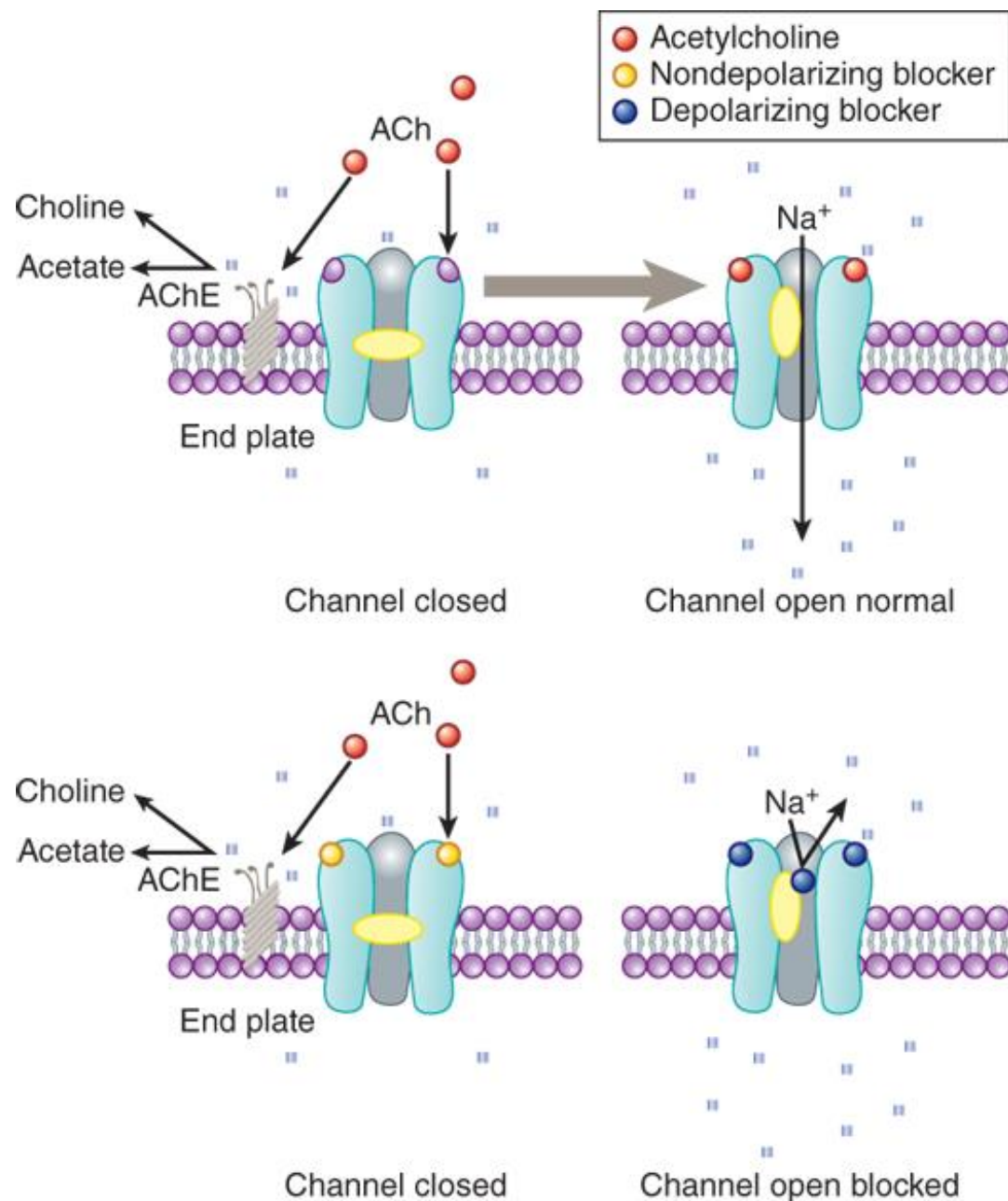
AMINOSTEROIDSIntermediate-acting:

Rocuronium	Biliary primarily, minor renal excretion	1-2 min	20-120 min	0.8
Vecuronium	Biliary (40-75%) and renal (~30%)	2.5-3 min	20-40 min (~45-65 min)	6

Pancuronium is no longer available in US or Canada. Long-acting; causes tachycardia due to sympathomimetic stimulation and cardiac muscarinic receptor blockade; high incidence of postop residual neuromuscular weakness

*Potency is compared to tubocurarine potency = 1.

Plasma clearance, volume of distribution, and elimination half-lives may be influenced by patient age, volatile anesthetics, and the presence of renal or hepatic disease.



Schematic diagram of the interactions of drugs with the acetylcholine receptor on the end plate channel (structures are purely symbolic).

Top: The action of the normal agonist, acetylcholine in opening the channel (red spheres labeled ACh).

Bottom, left: A nondepolarizing blocker, eg, rocuronium, is shown as preventing the opening of the channel when it binds to the receptor (depicted as small yellow spheres occupying active sites).

Larger doses of the nondepolarizing NMBAs → the drugs enter the channel pores → more intense motor blockade (not shown)

Bottom, right: A depolarizing blocker, eg, succinylcholine, both occupying the receptor and blocking the channel. Normal closure of the channel gate is prevented, and the blocker may move rapidly in and out of the pore (shown as blue spheres occupying active sites and inside the pore).

Depolarizing blockers may desensitize the end plate by occupying the receptor and causing persistent depolarization.

An additional effect of drugs on the end plate channel may occur through changes in the lipid environment surrounding the channel (not shown).

General anesthetics and alcohols may impair neuromuscular transmission by this mechanism.

Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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Effects: Sequence of Non-depolarizing NM Block

Initially, small, rapidly moving muscles, such as those of the eyes, jaw, and larynx relax.

Then, muscles of the limbs, trunk and abdomen

Then, the intercostal muscles

Finally, the diaphragm, and respiration ceases

Recovery of skeletal muscles usually occurs in the reverse order, so that the diaphragm is the first to regain normal function.

› These effects are monitored using a device that produces transdermal electrical stimulation of one of the peripheral nerves to the hand or face, and the evoked contractions (twitches) are recorded.

NMBAs: Skeletal Muscle Paralysis Therapeutic Use

Facilitate intubation

relax laryngeal and jaw muscles

Facilitate operative manipulation

- Adjuvant in surgical anesthesia
 - Allows lighter level of anesthesia
-

Mechanical ventilation

- Improves thoracic compliance in critically ill patients with ventilatory failure
-

Adjunct in orthopedic procedures

- Dislocations, fracture alignments
-

Consciousness and sensorium remain intact

- *no pain control, sedation, or amnestic effect* –

Not a substitute for adequate depth of anesthesia.

- Monitor with nerve stimulator to prevent over- and under-dosing

Reversal of nondepolarizing NMBA-induced muscle relaxation

Isoquinolines

Atracurium, Cisatracurium

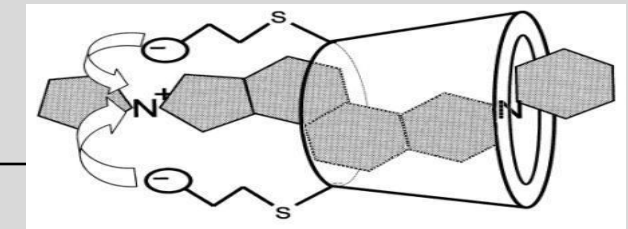
Cholinesterase inhibitor

- **neostigmine** or pyridostigmine
- Augments ACh concentration in the synapse to outcompete the nondepolarizing blocker

Neostigmine: faster onset of action than pyridostigmine, a shorter duration of action, and generally considered more effective.

Aminosteroids

Rocuronium, Vecuronium



Sugammadex

- Chelating agent forms a complex with steroid NMBA in a 1:1 ratio → rapid ↓NMBA in levels
- Complex is excreted in urine.
- Cholinesterase inhibitors are also effective reversal agents.

Note:

Cholinesterase inhibitors are administered with atropine or glycopyrrolate to prevent vagal bradycardia.

Large doses of cholinesterase inhibitor administered when neuromuscular block is minimal (near recovery) can result in neuromuscular dysfunction due to excessive acetylcholine in the neuromuscular junction.

Sugammadex adverse effects: Serious adverse effects are rare.

Bradycardia, ECG abnormalities (monitor during therapy)

May encapsulate steroid contraceptives and reduce effectiveness of the contraceptive.

May increase coagulation parameters; clinically significant bleeding has not been reported.

Prolonged apnea, risk factors

Inadequate reversal, residual NM block, prolonged surgery, opioids

Electrolyte abnormalities may potentiate block

Acidosis, hypokalemia, hypocalcemia, hypermagnesemia

Burns, Skeletal muscle trauma

Increased doses of NMBA required

Neuromuscular diseases enhance block

Myasthenia gravis: Autoantibodies directed against the NM nicotinic receptors

Lambert-Easton myasthenic syndrome: Autoantibodies directed against voltage-gated Ca^{2+} channels, which decreases release of ACh.

Age

Neonates may have increased sensitivity to block.

Older patients may have reduced hepatic / renal function.

Hypersensitivity reactions are rare

Drug interactions:

Aminoglycosides and other antibiotics increase ACh release or interfere with receptor function and prolong muscle weakness and respiratory depression

Many other drugs: Potentiate or inhibit NMBA effects, enhance respiratory depression, or contribute to electrolyte imbalance.

Comparing the nondepolarizing NMBAs

(The takeaways are in red.)

Atracurium	Cisatracurium	Rocuronium	Vecuronium
Hemodynamic effects: • Histamine release causes skin flushing, hypotension, tachycardia Metabolism: • plasma esterase hydrolysis and Hofmann elimination → laudanosine • high concentrations → accumulation (over days) → can cause CNS excitation and lower seizure • → Dosage adjustment of inhaled anesthetic may be required. No dose adjustment needed for liver or kidney impairment	Minimal hemodynamic effects: • Does not release histamine and not vagolytic Metabolized by Hofmann elimination • more potent → lower dose compared to atracurium → low accumulation of laudanosine • not likely to cause significant toxicity due to laudanosine No dosage adjustment for kidney or liver impairment necessary	Minimal hemodynamic effects: • Does not release histamine and not vagolytic Quick onset: • Least potent of the nondepolarizing NMBAs in clinical use → rapid onset • Often used for rapid sequence intubation Excreted in bile, minor excretion in urine • Suitable for patients with renal impairment • Dose adjustment may be needed in patients with liver disease	Minimal hemodynamic effects: • Does not release histamine and not vagolytic >6 times more potent than rocuronium • Slightly slower onset than rocuronium • not suitable for rapid sequence intubation Hepatic metabolism and excretion through both biliary and renal routes • Patients with liver or kidney impairment may have prolonged response

Pancuronium: No longer available in US or Canada. Long-acting; tachycardia due to sympathomimetic stimulation and cardiac muscarinic receptor blockade; High incidence of postop residual neuromuscular weakness

Depolarizing NMBA

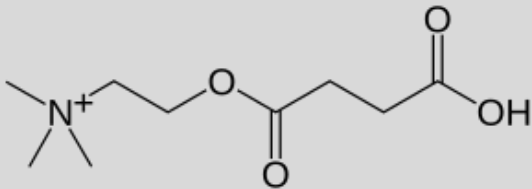
*Succinylcholine

Drug	Elimination	Onset	Duration	Potency
Succinylcholine	Plasma cholinesterase Only a small fraction of the dose reaches the motor end plate.	0.5 min	<8 min	0.4 tubocurarine = 1
	Pharmacogenomics: Duration is prolonged in patients with deficient or absent plasma cholinesterase. The <i>BCHE</i> genetic variants are rare.			

- Succinylcholine has a very short duration because it is rapidly hydrolyzed by liver butyrylcholinesterase and plasma cholinesterase (mainly plasma cholinesterase).
Only a small fraction of the dose reaches the neuromuscular junction.
- Succinylcholine is not degraded by AChE.
- **Succinylcholine blockade is terminated by diffusion** of the drug from the motor end plate into extracellular fluid.

Initial stimulation with muscle contractions	Phase I Block	Phase II Block	Later in Phase II Block
SCh binds nAChRs ↓ ion channels open ↓ membranes are depolarized ↓ small spontaneous twitches (fasciculations) as Ca ²⁺ is released from the sarcoplasmic reticulum	Depolarized membranes remain depolarized and unresponsive to subsequent impulses. Ca ²⁺ is taken up by the sarcoplasmic reticulum ↓ flaccid paralysis = skeletal muscle relaxation	• Initial end plate depolarization decreases. • Membrane becomes repolarized. <i>However,</i> • the membrane can't easily be depolarized again because it is desensitized. The channels behave as if they are in a prolonged closed state. (Mechanism is unclear.)	A non-sustained twitch response to a tetanic stimulus. The characteristics of the blockade are nearly identical to those of a nondepolarizing block.
Depolarizing block is augmented, not reversed, by cholinesterase inhibitors.		Prolonged exposure: large single dose, repeated dosing, continuous infusion	

Malignant hyperthermia Pediatric patients with undiagnosed skeletal muscle myopathy.	- Patients with congenital myopathies ➤ Volatile anesthetics can also cause malignant hyperthermia. - Abnormal release of Ca^{2+} stores from skeletal muscle - Hyperthermia, hyperkalemia, acidosis, hemo-/myoglobinuria, CV complication - Tx: dantrolene RyR1 blocker prevents Ca^{2+} release in skeletal muscle
Masseter spasm	Transient lower jaw muscle spasm may initially result in some difficulty opening the mouth (common in children; may be associated with malignant hyperthermia)
Myalgia Rhabdomyolysis Myoglobinuria	Muscle fasciculations damage skeletal muscle cells and release potassium into the blood stream (probable cause) .
Hyperkalemia	- Rapid loss of intracellular K^{+} (caution in burn patients, NM trauma, disease) - Can be life-threatening in patients with preexisting hyperkalemia
Cardiac dysrhythmia	- Ventricular arrhythmias resulting from hyperkalemic rhabdomyolysis - Bradycardia possibly due to muscarinic stimulation of SA node by metabolite succinylmonocholine - Atropine or glycopyrrolate pre-treatment to prevent bradycardia



Succinylcholine: Adverse Effects and Their Mechanisms

↑ Intra gastric pressure	• Muscle fasciculations may increase risk of regurgitation and aspiration
↑ Intraocular pressure	• Muscle fasciculations may induce contraction of extra-ocular muscles with distortion and compression of the globe
↑ Intracranial pressure	• Transient. May result from stimulation of afferent muscle activity leading to increased cerebral flow.
Plasma cholinesterase deficiency Prolonged apnea	• Congenital: Homozygous for atypical <i>BCHE</i> gene • Acquired: Burn patients, anemia, decompensated heart disease, infections, malignant tumors, severe hepatic dysfunction, pregnancy, others
Hypersensitivity reaction	• Avoid in patients with a history of allergic reaction to succinylcholine or any component of the formulation
CONTRAINDICATIONS	• h/o malignant hyperthermia; myopathy / muscle paralysis; rhabdomyolysis; multiple trauma; spinal cord or upper motor neuron injury; major burns; ocular injury → exaggerated hyperkalemic response

Check your knowledge

- **Mechanistically, what is the difference between depolarizing block and nondepolarizing block?**
- **What are the mechanisms of action of the neuromuscular blockade reversal agents for each NMBA subclass?**

Check your knowledge

- **Which nondepolarizing NMBA may cause prolonged duration of blockade in patients with liver or kidney impairment?**
- **Which NMBA can cause flushing and hypotension and what is the cause?**
- **Which NMBA can cause malignant hyperthermia in individuals with skeletal muscle disease?**
- **What is the effect of NMBA administration to patients with autoimmune diseases, myasthenia gravis and Lambert-Eaton myasthenic syndrome?**

Check your knowledge

- **What is the mechanism of muscle fasciculations caused by succinylcholine?**
- **List the serious adverse effects related to muscle fasciculations.**
- **What is a potential mechanism of the transient increased intracranial pressure caused by succinylcholine?**
- **What is the name of the byproduct of NMBA metabolism that in high concentrations is associated with lowering the seizure threshold?**

Check your knowledge

- **What drug(s) can be administered prior to succinylcholine to mitigate the bradycardia caused by the NMBA?**
- **What drugs are used for rapid sequence intubation and why?**
- **Succinylcholine is contraindicated in what conditions?**

Summary of Nicotinic Receptor Blockers:

- Ganglionic blockers
- Neuromuscular blockers

Ganglionic Blockers

- Ganglionic blockers inhibit the transmission of nerve impulses in the autonomic ganglia by acting as noncompetitive antagonists at neuronal nicotinic receptors. They suppress all parasympathetic and sympathetic outflow and autonomic homeostatic reflexes.
- Historically, they have been used to treat hypertension. Responses are complex and largely unpredictable and use is limited by side effects such as severe orthostatic hypotension, constipation, and urinary retention.
- Mecamylamine is an oral drug, crosses blood-brain barrier and inhibits nicotinic receptors in the CNS. Trimethaphan is parenteral only and does not cross the BBB. Hexamethonium is used in neuroscience research.

NMBAs

- Neuromuscular blocking agents (NMBAs) are highly polar compounds that cause temporary paralysis by blocking the transmission of nerve impulses to the muscles by acting on nicotinic receptors in the neuromuscular junction.
- NMBAs are used to induce muscle relaxation to facilitate intubation, surgical procedures in conjunction with anesthesia, and in mechanical ventilation.
- Nondepolarizing agents, atracurium, cisatracurium, rocuronium, and vecuronium, are competitive inhibitors of ACh. They have durations of action from minutes to longer than an hour. Prolonged paralysis that can lead to respiratory paralysis (apnea). Bradycardia, hypotension, and anaphylaxis are other potential serious effects.
- Succinylcholine (SCh) is a depolarizing NMBA with a rapid onset and short duration of action. It binds the ACh receptors, initially causing channel opening, membrane depolarization and spontaneous muscle twitching. It maintains the receptors in the depolarized state, which are unresponsive to subsequent activation – Phase I Block. With prolonged exposure, the receptors become repolarized but desensitized to activation by ACh – Phase II Block.
- Succinylcholine may cause malignant hyperthermia, myopathy, rhabdomyolysis, hyperkalemia and cardiac dysrhythmia, bradycardia, increased intragastric, intraocular, and intracranial pressure,
- Cholinesterase inhibitors reverse block by the isoquinolines and aminosteroids. Sugammadex reverses rocuronium and vecuronium block by complexing the drugs in plasma. Depolarizing block by SCh reverses spontaneously as the drug diffuses out of the synapse.

1. What is the overall effect of ganglion blockade by mecamylamine?

- Ganglion blockers suppress all autonomic outflow nAChRs of both parasympathetic and sympathetic ganglia and adrenal medulla, which suppresses all autonomic outflow, and suppress autonomic homeostatic reflexes that normally moderate autonomic responses: normal cardiac, vascular, neurologic, muscle, and neurohumoral reflexes

2. What are the mechanism of orthostatic hypotension and symptoms caused by mecamylamine?

- Orthostatic hypotension (a significant drop in blood pressure on standing) and syncope is pronounced because postural reflexes (sympathetic responses) that normally prevent venous pooling are abolished. Symptoms are dizziness, lightheadedness, and fainting (syncope).

3. Why should mecamylamine be prescribed only for patients who are reliable and cooperative with the therapy?

- Mecamylamine dosing can be complicated, therapy must be carefully monitored, and the drugs cause orthostatic hypotension. Patients should be cautioned about all potential side effects and drug interactions.

4. Since mecamylamine is rarely used, why is it important for students to learn about it (other than because it might appear on a board exam)?

- Understanding the effects of ganglion blockers requires a deep knowledge of the autonomic nervous system (ANS). Grasping the concepts means students can appreciate how ganglion blockers influence various physiological processes and manage the potential side effects and clinical implications effectively.

Check your knowledge

- **Mechanistically, what is the difference between depolarizing block and nondepolarizing block?**
- In depolarizing block, the drug initially causes membrane depolarization, ion channel opening, and activation, followed by depolarization but unresponsive to subsequent impulses.
- Competitive inhibitors of acetylcholine at the binding pockets prevent channel opening.
- **What are the mechanisms of action of the neuromuscular blockade reversal agents for each NMBA subclass?**
- Isoquinolines: Cholinesterase inhibitors enhance synaptic concentrations of acetylcholine. The neurotransmitter outcompetes the nondepolarizing NMBAs facilitating transmission of impulses across the neuromuscular junction.
- Aminosteroids: Either cholinesterase inhibitors or sugammadex. Sugammadex is a gamma-cyclodextrin that complexes (encapsulates) the aminosteroid in plasma in a 1:1 ratio causing a rapid decrease in NMBA plasma concentrations. The complex is excreted in the urine.
- Succinylcholine diffuses out of the synapse, terminating neuromuscular blockade.

Check your knowledge

- **Which nondepolarizing NMBA may cause prolonged duration of blockade in patients with liver or kidney impairment?**
- Vecuronium is metabolized in the liver and excreted in bile and urine. Impairment of the organs increases the duration of blockade.
- **Which NMBA can cause flushing and hypotension and what is the cause?**
- Atracurium releases histamine, which acts on vascular histamine receptors, causing vasodilation, thus, a drop in blood pressure (hypotension) and skin reddening (flushing).
- **Which NMBA can cause malignant hyperthermia in individuals with skeletal muscle disease?**
- Succinylcholine and also inhaled anesthetics.
- **What is the effect of NMBA administration to patients with autoimmune diseases, myasthenia gravis and Lambert-Eaton myasthenic syndrome?**
- MG is characterized by antibodies directed against neuromuscular nicotinic receptors. LEMS is characterized by antibodies directed against presynaptic voltage-gated calcium channels, which reduces ACh release from presynaptic terminals. Both diseases cause potentiation of neuromuscular blockade by NMBAs.

Check your knowledge

- **What is the mechanism of muscle fasciculations caused by succinylcholine?**
- Succinylcholine binds acetylcholine receptors initially causing ion channel opening and depolarization of membranes → small spontaneous twitching.
- **List the serious adverse effects related to muscle fasciculations.**
- Rhabdomyolysis, hyperkalemia → cardiac dysrhythmia, increased intragastric and intraocular pressure.
- **What is the likely mechanism of the transient increased intracranial pressure caused by succinylcholine?**
- Stimulation of muscle activity by succinylcholine may stimulate afferent muscle activity leading to transient increased cerebral blood flow, thus transient increased intracranial pressure.
- **What is the name of the byproduct of NMBA metabolism that in high concentrations is associated with lowering the seizure threshold?**
- Atracurium is metabolized in part by nonbiologic Hofmann elimination, which produces laudanosine. High laudanosine concentration in the CNS (such days of therapy in ICU) may cause seizures (demonstrated in animal studies)

Check your knowledge

- **What drug(s) can be administered prior to succinylcholine to mitigate the bradycardia caused by the NMBA?**
- Antimuscarinic drugs atropine and glycopyrrolate. They also decrease secretions.
- Bradycardia is attributed to the succinylcholine metabolite, succinylmonocholine.
- **What drugs are used for rapid sequence intubation and why?**
- Succinylcholine and rocuronium both have a rapid onset of action, suitable for RSI.
- **Succinylcholine is contraindicated in what conditions?**
- Patients with a h/o malignant hyperthermia; myopathy / muscle paralysis; rhabdomyolysis; multiple trauma; spinal cord or upper motor neuron injury; major burns; ocular injury → exaggerated hyperkalemic response

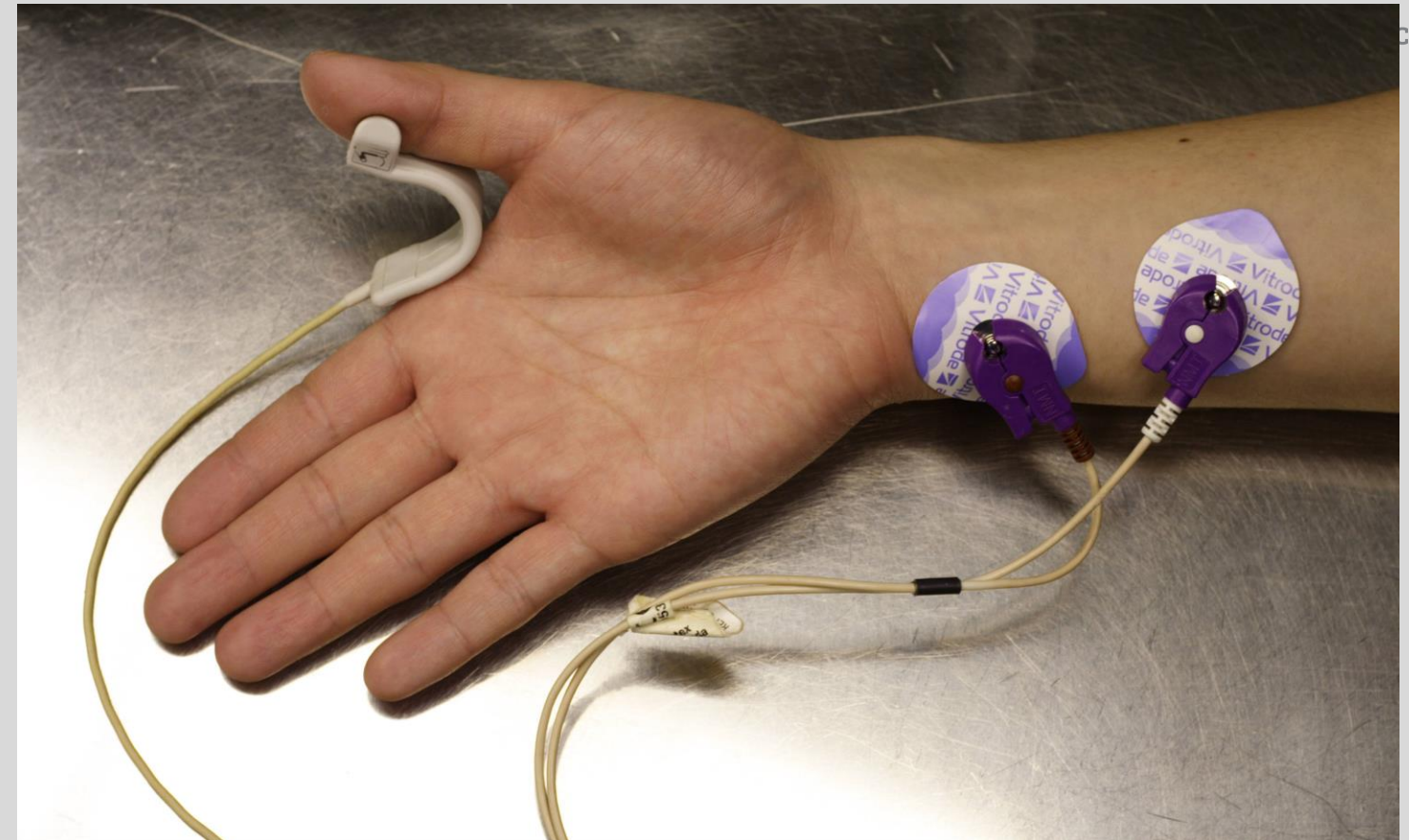
Endotracheal intubation while wearing powered air-purifying respirators



Two anesthesiologists using a videolaryngoscope for endotracheal intubation while wearing powered air-purifying respirators.

From: Hong-Fei Z, Lu-Long B, Lin Y, et al. Response of Chinese anesthesiologists to the COVID-19 outbreak. *Anesthesiology* 2020. DOI: [10.1097/ALN.0000000000003300](https://doi.org/10.1097/ALN.0000000000003300). Copyright © 2020 the American Society of Anesthesiologists. Reproduced with permission from Wolters Kluwer Health. Unauthorized reproduction of this material is prohibited.

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Neuromuscular transmission (NMT) monitor for measuring neuromuscular blockade measures response of muscles to nerve stimulation to ensure appropriate dosing of NMBAs and that patients recover adequately before leaving the operating room.

https://commons.wikimedia.org/wiki/File:NMT_monitor_for_measuring_neuromuscular_blockade.jpg author ignis

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